On Gomory Cutting Planes

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Abstract. We study Gomory's cutting-plane method [Gom58, Gom60, Gom63] from the viewpoint of integer linear programming. We start with an LP relaxation and ask how to derive linear inequalities that every integer feasible point satisfies but a fractional optimum may violate. We use Chvátal-Gomory cuts to formalize this rounding step, prove their validity for the integer hull, and derive Gomory's fractional cut from the optimal simplex tableau. We then explain finite convergence through Chvátal rank [Chv73, Sch80] and separate this structural theorem from the harder problem of choosing useful cuts efficiently. Finally, we connect the method to polyhedral combinatorics, lift-and-project systems, and CP proof complexity, and we record open problems on rank bounds, TSP relaxations, matching polytopes, and cut-selection rules.

1 Introduction

Integer programming grew out of two developments that at first sight pull in opposite directions. On one side, the simplex method and the polyhedral theory of linear programming made continuous optimization computationally effective and geometrically transparent [Dan51, Dan63, Chv83, Sch86, NW88]. On the other side, combinatorial optimization produced natural problems whose feasible solutions were discrete objects, such as matchings, tours, flows with integrality requirements, and set systems [DFJ54, Edm65b, Edm65a, HK70]. The tension between these two viewpoints is exactly the tension that cutting planes exploit. We solve a continuous relaxation, but we do not accept the relaxation as the problem; instead, we ask which linear consequences of integrality can be recovered from it.

Gomory's original papers introduced this idea in algorithmic form [Gom58, Gom60, Gom63]. The tableau was not only a data structure for the simplex method. It became a certificate of which arithmetical obstruction remained at the current LP optimum. Chvátal then abstracted the rounding step into the closure operation now called the Chvátal-Gomory closure [Chv73], and Schrijver developed the systematic polyhedral theory of cutting planes and rank [Sch80, Sch86]. From this point of view, Gomory cuts are one concrete rule inside a broader proof system for deriving valid inequalities of a rational integer hull.

The same language became central in combinatorial optimization. Edmonds' matching polytope theorem showed that polynomial-time algorithms may be hidden inside exponentially large linear

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descriptions, provided the relevant inequalities can be separated efficiently [Edm65b, Edm65a, GLS81, GLS88]. The subtour relaxation for the TSP and the Held-Karp bound gave another canonical testing ground, where a natural LP relaxation is useful but not integral [DFJ54, HK70, Chr76, Ser78]. These examples explain why rank questions are subtle. A relaxation may be algorithmically useful, separable, and geometrically meaningful without having small rank under a particular closure operation.

The modern theory enlarged the family of closures. Disjunctive programming and lift-and-project methods replace a fractional point by a higher-dimensional relaxation whose projection contains stronger valid inequalities [Bal71, NW88, BCC93]. The Sherali-Adams, Lovász-Schrijver, and Lasserre hierarchies made this principle systematic, connecting integer programming to semidefinite programming, sums of squares, and proof complexity [SA90, LS91, Las01]. Split cuts and mixed-integer Gomory cuts then became the practical workhorses of branch-and-cut implementations [Gom63, CKS90, CCZ14]. Thus, the subject sits between two questions that should be kept separate throughout this note. One question asks which inequalities are valid. The other asks which valid inequalities can be found, represented, and used efficiently.

There is also a proof-complexity reading of the same material. A cutting-plane derivation is a proof in which each line is a linear inequality and each inference is an arithmetical rounding step [CKS90]. This places cutting planes alongside Resolution, Frege, and other propositional proof systems [Hak85, Bus87, Kra95, BSW01]. Lower bounds for cutting-plane proofs, beginning with Cook, Coullard, and Turán and continuing through work of Pudlák and others, show that the ability to count with integer coefficients is powerful but still limited [CKS90, Pud97]. This is one reason the open problems in § 8 are not only algorithmic. They ask how much discrete structure a linear proof can reveal before the proof itself becomes too long.

2 Integer Programming and LP Relaxations

The difficulty of integer programming is geometric in a very literal sense. Linear programming is designed to optimize over convex sets, whereas the feasible set of an integer program is a discrete subset of Euclidean space. The standard relaxation therefore replaces the arithmetical requirement $\mathbf{x} \in \mathbb{Z}^n$ by a convex feasible region P, solves a problem that is now tractable, and then faces the question of how much information was lost by forgetting integrality [Sch86, NW88, CCZ14].

Cutting planes answer this question by refusing to branch on possible integer assignments. Instead, we keep the convex language of linear inequalities and ask whether the current relaxation contains fractional points that can be separated from every integer feasible point by a new inequality. Thus, the method is neither purely combinatorial nor purely continuous; it is a procedure for translating integrality into convex geometry one inequality at a time.

2.1 Setup and Notation

Let $\mathbf{A} \in \mathbb{Q}^{m\times n}$, $\mathbf{b} \in \mathbb{Q}^m$, and $\mathbf{c} \in \mathbb{Q}^n$. We write $[n] := \{1, \dots, n\}$. The LP relaxation is the linear program

$$\mathsf{LP}: \quad z_{\mathsf{LP}} := \min \mathbf{c}^{\mathsf{T}}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} \le \mathbf{b}, \quad \mathbf{x} \ge 0, \tag{2.1}$$

with feasible polytope $P := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge 0\}$. The corresponding integer program is obtained by imposing $\mathbf{x} \in \mathbb{Z}^n$.

$$\mathsf{IP}: \quad z_{\mathsf{IP}} := \min \mathbf{c}^{\mathsf{T}}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} \le \mathbf{b}, \quad \mathbf{x} \ge 0, \quad \mathbf{x} \in \mathbb{Z}^n. \tag{2.2}$$

Definition 2.1 (Integer hull)**.** The integer hull of P is $P_{\mathrm{I}} := \operatorname{conv}(P \cap \mathbb{Z}^n)$, the convex hull of all integer points in P. ⌟

Since $P_{\mathrm{I}} \subseteq P$, the optima satisfy $z_{\mathsf{LP}} \le z_{\mathsf{IP}}$ for minimization. The integrality gap $z_{\mathsf{IP}}/z_{\mathsf{LP}}$, when both quantities are positive, measures how far the relaxation may stray from the integer optimum. Optimizing over $P \cap \mathbb{Z}^n$ is the same as optimizing over P_{I}, because a linear function attains its minimum over the convex hull at an extreme point and the extreme points of P_{I} are convex combinations of feasible integer points only in the trivial way.

The cutting-plane method exploits this equivalence without first writing down all facets of P_{I}. At a fractional LP optimum, the current relaxation has already told us where it is too large. We then seek a linear inequality that is valid for all of P_{I} and violated by that fractional optimum, thereby removing the current error while preserving every genuine integer solution.

2.2 The Integer Hull as a Polytope

Before attempting to describe P_{I} by linear inequalities, we should first justify that such a finite description exists. If P is bounded, the set $P \cap \mathbb{Z}^n$ is finite and there is no mystery. If P is an unbounded rational polyhedron, however, the integer feasible set may be infinite, and the assertion that its convex hull remains polyhedral becomes a genuine theorem.

Theorem 2.2 (Meyer [Mey74])**.** If P is a rational polyhedron, then P_{I} is a rational polyhedron. In particular, if P is a rational polytope, then P_{I} is a rational polytope. ⌟

Proof (Proof sketch)*.* We recall the bounded case, which is the only case needed for the geometric intuition in this note. If P is a bounded polytope, then $P \cap \mathbb{Z}^n$ is finite, because it lies in a box $[-M, M]^n$ for some real M and hence contains at most $(2\lceil M \rceil + 1)^n$ integer points. The convex hull of finitely many points is a polytope, and all its vertices are among the finitely many integer points used to form the hull. The full rational-polyhedron statement is Meyer's theorem; its proof decomposes the recession cone and the bounded part of a rational polyhedron, showing that the integer hull has a finite rational generating description even when $P \cap \mathbb{Z}^n$ is infinite. □

Theorem 2.2 guarantees that P_{I} has a finite linear description, but the theorem should not be mistaken for an algorithmic solution. The description may be far larger than the original relaxation. For example, Edmonds' description of the matching polytope contains an odd-set inequality for every odd subset of vertices, and the complete graph therefore gives exponentially many such inequalities [Edm65b, Edm65a]. The cutting-plane method avoids writing down all of them in advance. It searches for inequalities only when the LP relaxation exhibits a fractional optimum that needs to be removed.

3 Chvátal-Gomory Derivations

The Chvátal-Gomory framework answers the first formal question raised by the cutting-plane program. Suppose that a linear inequality is valid for the relaxation P, and suppose further that its left-hand side takes integer values on every integer point. Then a fractional right-hand side contains slack that no integer feasible point can use, and we may round it down without losing any integer solution. This elementary rounding step is the basic inference rule behind CG cuts.

There are two equivalent ways to present the rule. One may start with an arbitrary valid inequality for P whose coefficient vector is integral, or one may obtain such an inequality by taking a nonnegative linear combination of the displayed constraints defining P. The second form is closer to algorithms, because it explains how a cut is derived from the current relaxation, while the first form is cleaner for defining the closure.

3.1 Definition and Validity

Definition 3.1 (CG cut)**.** Let $\lambda \geq 0$ be multipliers such that $\mathbf{a} := \lambda^\mathsf{T}\mathbf{A} \in \mathbb{Z}^n$. The valid inequality $\mathbf{a}^\mathsf{T}\mathbf{x} \leq \lambda^\mathsf{T}\mathbf{b}$ holds for all $\mathbf{x} \in P$. The Chvátal-Gomory cut derived from λ is

$$\mathbf{a}^\mathsf{T}\mathbf{x} \leq \left\lfloor \lambda^\mathsf{T}\mathbf{b} \right\rfloor .$$

⌟

Theorem 3.2 (Validity of CG cuts)**.** Every CG cut is valid for P_I. ⌟

Proof. The proof isolates the one place where integrality is used. A real point satisfying $\mathbf{a}^\mathsf{T}\mathbf{x} \leq \beta$ need not satisfy $\mathbf{a}^\mathsf{T}\mathbf{x} \leq \lfloor\beta\rfloor$, but an integer point does satisfy the rounded inequality whenever $\mathbf{a}$ is integral, because the left-hand side is itself an integer.

Let $\mathbf{z} \in P \cap \mathbb{Z}^n$. Since $\lambda \geq 0$ and $\mathbf{Az} \leq \mathbf{b}$ hold componentwise, we obtain

$$\mathbf{a}^\mathsf{T}\mathbf{z} = \lambda^\mathsf{T}\mathbf{Az} \leq \lambda^\mathsf{T}\mathbf{b}.$$

The vector $\mathbf{a}$ is integral by assumption, and $\mathbf{z}$ is integral by choice, so $\mathbf{a}^\mathsf{T}\mathbf{z} \in \mathbb{Z}$. An integer that is at most $\lambda^\mathsf{T}\mathbf{b}$ is also at most $\left\lfloor \lambda^\mathsf{T}\mathbf{b} \right\rfloor$, whence

$$\mathbf{a}^\mathsf{T}\mathbf{z} \leq \left\lfloor \lambda^\mathsf{T}\mathbf{b} \right\rfloor .$$

Thus, the CG cut holds for every point in $P \cap \mathbb{Z}^n$. Since a linear inequality that holds on a set also holds on its convex hull, the same inequality holds on $P_\mathrm{I} = \mathrm{conv}(P \cap \mathbb{Z}^n)$. □

The proof is almost tautological, and this is the point. The power of the CG step lies entirely in the choice of λ. When $\lambda = \mathbf{e}_i$, the CG step applied to the i-th constraint yields $\mathbf{a}_i^\mathsf{T}\mathbf{x} \leq \lfloor b_i \rfloor$, which tightens the constraint whenever $b_i \notin \mathbb{Z}$. The non-trivial case arises when a combination of constraints produces cancellation, yielding an integral coefficient vector not present in any single row of $\mathbf{A}$. The right-hand side of the combined constraint is then strictly smaller than any right-hand side in the original system, and rounding down tightens it further. Gomory's insight is that the simplex tableau at optimality reveals exactly which combination to use.

3.2 Chvátal Closures and Rank

Applying all possible CG cuts simultaneously yields the Chvátal closure. This is the tightest set obtainable by one round of CG inference, and iterating the closure operation converges to P_I.

Definition 3.3 (Chvátal closure and rank)**.** The Chvátal closure P' of P is the intersection of all halfspaces $\{\mathbf{a}^\mathsf{T}\mathbf{x} \leq \lfloor\beta\rfloor\}$ where $\mathbf{a}^\mathsf{T}\mathbf{x} \leq \beta$ ranges over all valid inequalities for P with $\mathbf{a} \in \mathbb{Z}^n$. Define

$P^{(0)} := P$ and $P^{(k+1)} := (P^{(k)})'$ inductively. The Chvátal rank $r(P)$ is the smallest $r \geq 0$ such that $P^{(r)} = P_{\mathrm{I}}$. ⌟

Since every CG cut is valid for P_{I} by Theorem 3.2, we always have $P_{\mathrm{I}} \subseteq P^{(k)}$ for every k. The sequence $P = P^{(0)} \supseteq P^{(1)} \supseteq \cdots$ descends toward P_{I} from above; the question is whether it reaches P_{I} in finitely many steps.

Theorem 3.4 (Chvátal [Chv73]; Schrijver [Sch80])**.** Every rational polytope has finite Chvátal rank. ⌟

Proof (Proof sketch). The full proof is a structural theorem about rational polyhedra, not a tableau calculation. The main invariant is that each closure round removes all points that violate a rounded valid inequality whose coefficients are integral, and rationality ensures that the possible denominators of the relevant faces can be controlled. Chvátal proved finite convergence for rational polytopes, and Schrijver extended and sharpened the polyhedral argument [Chv73, Sch80].

In the bounded case, one may view the proof as a finite descent on rational faces. Clearing denominators gives an integral description of the affine span of each face, and every non-integral face of the current relaxation admits an integral valid inequality whose rounded version cuts that face away while preserving the integer hull. Since a rational polytope has finitely many faces at each stage and the face lattice strictly changes whenever such a face is removed, repeated closure cannot avoid the integer hull forever. The general proof makes this descent quantitative through the rational data and the recession cone, which is why the theorem applies to rational polyhedra as well. □

Theorem 3.4 is qualitative. It proves that the closure process eventually reaches the integer hull, but it gives no useful guarantee that the number of rounds is small. This distinction is crucial. Cutting planes provide a complete proof system for integer validity, whereas efficient optimization requires good bounds on rank and good rules for selecting useful cuts. We return to this gap in § 8.

4 The Gomory Fractional Cut

The Chvátal framework tells us what kind of inequality is valid, but it does not tell us which inequality to try at a given fractional optimum. Gomory's contribution is precisely this algorithmic step. Once the simplex method has produced an optimal basis, each row of the tableau expresses one basic variable as an affine function of the non-basic variables. If the current value of that basic variable is fractional, while the variable itself must be integral in every feasible integer solution, then the row contains an arithmetical contradiction waiting to be exposed.

The resulting cut uses only the fractional parts of that row. Intuitively, the current LP optimum sets every non-basic variable to zero and therefore leaves the fractional basic value unchanged. Any integer feasible solution must move some non-basic variables enough to cancel this fractional residue. The Gomory cut records exactly this necessary movement.

4.1 The Optimal Simplex Tableau

Introduce slack variables to convert $\mathbf{Ax} \leq \mathbf{b}$ into $\mathbf{Ax} + \mathbf{s} = \mathbf{b}$ with $\mathbf{s} \geq 0$. A basis $\mathbf{B}$ is a set of m variables, original or slack, whose corresponding columns form an invertible $m \times m$ matrix; call that matrix $\mathbf{B}$ as well by standard abuse of notation. The remaining variables form the non-basic set N.

At the LP optimum, the basic variables satisfy $\mathbf{x_B} = \mathbf{B}^{-1}\mathbf{b} =: \bar{\mathbf{b}}$ and $\mathbf{x}_N = 0$. Expressing each basic variable in terms of the non-basic ones gives the optimal tableau

$$x_{B_i} = \bar{b}_i - \sum_{j \in N} \bar{a}_{ij}\, x_j, \qquad i = 1, \dots, m, \tag{4.1}$$

where $\bar{\mathbf{A}}_N := \mathbf{B}^{-1}\mathbf{A}_N$ contains the tableau coefficients. Primal feasibility requires $\bar{b}_i \geq 0$ for all i. If every $\bar{b}_i$ is integral, the LP optimum is also the IP optimum. Otherwise, we pick any row i with $\bar{b}_i \notin \mathbb{Z}$ and derive a cut from it.

4.2 Deriving the Cut

Write the fractional decompositions

$$\bar{b}_i = \left\lfloor \bar{b}_i \right\rfloor + f_i, \qquad f_i := \{\bar{b}_i\} \in (0, 1), \tag{4.2}$$

For each coefficient we also write

$$\bar{a}_{ij} = \lfloor \bar{a}_{ij} \rfloor + f_{ij}, \qquad f_{ij} := \{\bar{a}_{ij}\} \in [0, 1).$$

Here $\{t\} := t - \lfloor t \rfloor$ denotes the fractional part function. For negative t, say $t = -1/3$, we have $\{t\} = -1/3 - (-1) = 2/3$, so the fractional part is always non-negative.

Definition 4.1 (Gomory fractional cut)**.** Given an optimal tableau row i with $\bar{b}_i \notin \mathbb{Z}$, the Gomory fractional cut derived from row i is

$$\sum_{j \in N} f_{ij}\, x_j \geq f_i. \tag{4.3}$$

⌟

The cut separates the current LP optimum immediately. At the LP optimum, $x_j = 0$ for all $j \in N$, so the left-hand side equals 0, while $f_i > 0$. The cut therefore strictly separates the LP optimum from P_{I}.

4.3 Validity

Theorem 4.2 (Gomory cut validity)**.** The Gomory fractional cut $\sum_{j \in N} f_{ij}\, x_j \geq f_i$ is valid for P_{I}. ⌟

Proof. The proof has the same structure as the validity proof for CG cuts, but written directly in tableau coordinates. We split the tableau row into its integral and fractional parts. The integral part evaluates to an integer at every integer feasible solution, and therefore the remaining fractional expression must also be an integer; since this integer is strictly less than one, it must be nonpositive.

Let $\mathbf{x}$ be any integer-feasible point in P. Since the problem has rational data and $\mathbf{x} \in \mathbb{Z}^n$, both x_{B_i} and each x_j are integers. Substituting the fractional decompositions into the i-th tableau row gives

$$\begin{aligned} x_{B_i} &= \bar{b}_i - \sum_{j \in N} \bar{a}_{ij}\, x_j \\ &= \left(\left\lfloor \bar{b}_i \right\rfloor + f_i\right) - \sum_{j \in N} \left(\lfloor \bar{a}_{ij} \rfloor + f_{ij}\right) x_j. \end{aligned}$$

Rearranging to isolate the fractional parts gives

$$\underbrace{x_{B_i} - \left\lfloor \bar{b}_i \right\rfloor + \sum_{j \in N} \lfloor \bar{a}_{ij} \rfloor \, x_j}_{=:\, M} \;=\; f_i - \sum_{j \in N} f_{ij}\, x_j. \tag{4.4}$$

The quantity M is an integer, because $x_{B_i} \in \mathbb{Z}$, $\left\lfloor \bar{b}_i \right\rfloor \in \mathbb{Z}$, and $\lfloor \bar{a}_{ij} \rfloor \, x_j \in \mathbb{Z}$ for each j. Therefore, the right-hand side of Equation (4.4) is also an integer. Since $f_{ij} \geq 0$ and $x_j \geq 0$, we have

$$f_i - \sum_{j \in N} f_{ij}\, x_j \leq f_i < 1.$$

The right-hand side is an integer strictly less than 1, so it is at most 0, and therefore

$$\sum_{j \in N} f_{ij}\, x_j \geq f_i. \qquad \square$$

The structure of this proof deserves attention. The key identity (4.4) splits the tableau row into an integer part M and a fractional part. The constraint $0 \leq f_i < 1$ forces $M \leq 0$, which is the content of the cut. The integer parts $\lfloor \bar{a}_{ij} \rfloor$ contribute only to M and play no role in the cut coefficients; the cut is determined entirely by the fractional parts f_{ij} of the tableau row.

Remark 4.3 (Geometric interpretation). The tableau row i implicitly encodes the non-negativity constraint $x_{B_i} \geq 0$ in terms of the non-basic variables. The fractional parts f_{ij} measure the extent to which increasing each non-basic variable x_j forces x_{B_i} toward the next integer from above. The cut $\sum_j f_{ij} x_j \geq f_i$ is the weakest constraint, among those with coefficients equal to fractional parts of row i, that still separates $\mathbf{x}^*$ from all integer-feasible points satisfying $x_{B_i} \in \mathbb{Z}$. ⌟

Remark 4.4 (CG interpretation). Every Gomory cut is a CG cut. The cut arises by taking the constraint $x_{B_i} \geq 0$, written as $\sum_{j \in N} \bar{a}_{ij} x_j \leq \bar{b}_i$, together with the non-negativity constraints $x_j \geq 0$ multiplied by the integer parts $\lfloor \bar{a}_{ij} \rfloor$; the combined system has integral left-hand side coefficients, and the CG step yields (4.3). See Schrijver [Sch86] for the full derivation. Validity therefore also follows from Theorem 3.2. ⌟

5 The Cutting Plane Algorithm

The algorithm alternates between continuous optimization and arithmetical correction. We solve the current LP relaxation, inspect an optimal tableau, and add a cut only if the tableau certifies that the current optimum cannot be an integer optimum. Thus, the simplex method supplies the point to be removed, while the integrality argument supplies the inequality that removes it.

There is a subtle distinction between the mathematical closure process and a concrete implementation. The Chvátal closure adds all CG cuts of one round at once, whereas Gomory's algorithm adds selected tableau cuts one at a time. Finite Chvátal rank explains why the family of CG cuts is complete, but finite termination of an actual tableau algorithm also depends on the pivoting and cut-selection discipline. Classical finite-convergence proofs for Gomory's method impose such a discipline, for example by using lexicographic pivoting to avoid cycling.

5.1 The Algorithm

(1) Solve the LP relaxation. If the problem is infeasible, the IP is infeasible; stop. If unbounded, the IP is unbounded; stop. Otherwise, obtain an optimal basis $\mathbf{B}$ and LP optimal $\mathbf{x}^* = \bar{\mathbf{b}} \geq 0$.
(2) Check integrality. If $\bar{b}_i \in \mathbb{Z}$ for all i, then $\mathbf{x}^*$ is integer-optimal; stop and return $\mathbf{x}^*$.
(3) Derive a Gomory cut. Pick any i with $\bar{b}_i \notin \mathbb{Z}$. Form the fractional parts f_i and f_{ij} as in § 4. Add the constraint $\sum_{j \in N} f_{ij} x_j \geq f_i$ to the LP.
(4) Re-solve. Return to step 1.

5.2 Maintaining Feasibility

Every Gomory cut is valid for P_{I} by Theorem 4.2, so the integer optimum remains feasible throughout; the algorithm never excludes integer-feasible points. The LP feasible set strictly shrinks at each step because the current LP optimum violates the new cut.

In practice, one does not re-solve the LP from scratch after each cut. The current basis remains dual-feasible (the reduced costs retain the correct sign) but becomes primal-infeasible because the new cut is violated by $\mathbf{x}^*$. The dual simplex method restores primal feasibility while maintaining dual feasibility, typically in a small number of pivots. The key observation is that the new cut introduces exactly one additional row, and the dual simplex pivots until that row's slack becomes non-negative. For large instances, practitioners add several cuts per iteration before re-solving, amortising the re-solve cost over multiple cuts simultaneously.

6 Termination

Theorem 6.1 (Gomory [Gom58])**.** For a pure integer linear program with rational data, Gomory's fractional cutting-plane algorithm, equipped with an anti-cycling pivoting rule such as lexicographic dual simplex, terminates in finitely many iterations. ⌟

6.1 Why the Algorithm Must Terminate

The finite-termination proof should be read as a compatibility statement between two viewpoints. The polyhedral viewpoint says that CG cuts are complete for rational integer hulls. The tableau viewpoint says that, whenever the current optimal basis is fractional, the simplex tableau produces a CG cut that removes that particular basis solution. The anti-cycling rule supplies the missing algorithmic ingredient by ensuring that the method does not return indefinitely to equivalent fractional bases.

6.1.1 Each cut is valid and removes the current basis..

By Theorem 4.2, each added tableau cut is valid for P_{I}. The current basic feasible solution violates the cut because all non-basic variables are zero and the right-hand side f_i is positive. Hence, each successful iteration preserves all integer feasible points and removes the current fractional LP optimum.

6.1.2 Denominators are bounded..

All entries $\bar{b}_i = (\mathbf{B}^{-1}\mathbf{b})_i$ are rational. Cramer's rule gives $|\det(\mathbf{B})|$ as the common denominator of all entries of $\mathbf{B}^{-1}$. Since $\mathbf{B}$ is a submatrix of $[\mathbf{A} \mid \mathbf{I}]$ with rational entries having denominator at most Δ (the LCM of denominators in $\mathbf{A}$ and $\mathbf{b}$), Hadamard's inequality gives $|\det(\mathbf{B})| \leq \Delta^m \cdot m!$. The fractional part $f_i = \{\bar{b}_i\}$ is therefore a rational number with bounded denominator whenever the basis matrix is drawn from a bounded rational encoding of the current relaxation. This bounded-denominator phenomenon is one reason lexicographic finite-convergence arguments can rank tableaux by a well-founded measure rather than by objective value alone.

6.1.3 The closure theorem explains completeness..

Since every Gomory cut is a CG cut, Gomory's method operates inside the same inferential universe as the Chvátal closure. By Theorem 3.4, sufficiently many rounds of all CG cuts reach P_{I}. The classical tableau proof of Gomory's finite convergence should therefore be understood as the algorithmic refinement of this completeness theorem, showing that the particular cuts generated from fractional optimal tableaux, together with lexicographic pivoting, cannot miss the integer hull forever.

6.2 Complexity Caveats

Termination is not the same as efficiency. Several remarks clarify the computational status of the method.

Remark 6.2 (Exponential worst case). The number of Gomory cuts required can be exponential in the input size. Cook, Coullard, and Turán [CKS90] proved that establishing infeasibility of certain 0-1 integer programs requires exponentially many cutting plane steps, even with an optimal choice of cuts. No deterministic cut-selection strategy makes the method polynomial in general. ⌟

Remark 6.3 (Fixed dimension). For fixed n, Lenstra [Len83] showed that IP is polynomial in the number of constraints and the magnitude of the data, using the geometry of numbers (specifically, the LLL lattice-reduction algorithm). In fixed dimension, the integer hull has polynomially many vertices, and one can enumerate them without cutting planes. The cutting plane approach is therefore most relevant for large n and for structured instances where the cuts arise naturally from the problem. ⌟

Remark 6.4 (Total unimodularity). When $\mathbf{A}$ is totally unimodular, meaning that every square submatrix has determinant 0, $+1$, or -1, every vertex of P is integral and no cuts are needed. The LP optimal is also the IP optimal. Network flow matrices and incidence matrices of bipartite graphs are the canonical examples. Total unimodularity is the structural condition under which LP and IP coincide exactly; understanding why it fails for general combinatorial polytopes motivates the entire cutting plane program [HK56, GH62, Sch86]. ⌟

6.3 Mixed-Integer Generalization

Gomory [Gom63] extended the method to mixed-integer programs, where only some variables are required to be integer. The Gomory mixed-integer cut, or GMI cut, is derived from a tableau row exactly as before, except that the coefficients of continuous (non-integer) variables are treated differently from those of integer variables. Specifically, for a row i with $f_i > 0$ and non-basic variables partitioned into integer set $\mathcal{I}$ and continuous set $\mathcal{C}$, the GMI cut takes the form

$$\sum_{\substack{j\in\mathcal{I}\\ f_{ij}\le f_i}} f_{ij}\,x_j + \frac{f_i}{1-f_i}\sum_{\substack{j\in\mathcal{I}\\ f_{ij}>f_i}}(1-f_{ij})\,x_j + \sum_{j\in\mathcal{C},\,\bar a_{ij}>0}\bar a_{ij}\,x_j - \frac{f_i}{1-f_i}\sum_{j\in\mathcal{C},\,\bar a_{ij}<0}\bar a_{ij}\,x_j \ge f_i.$$

GMI cuts are the standard workhorse of modern branch-and-cut solvers. Their validity follows by the same integrality argument as Theorem 4.2, applied only to the integer non-basic variables.

7 Further Reading and Modern Context

The material above gives the classical core of Gomory's method, but the surrounding literature is much broader. The first major extension is the group-theoretic theory of valid inequalities initiated by Gomory and Johnson [GJ72a, GJ72b, Joh74]. This line studies cut-generating functions rather than individual tableau rows, and it explains why some inequalities recur across many mixed-integer models. Modern surveys and monographs place this theory together with split cuts, mixed-integer rounding cuts, and lattice-free convex sets [Cor08, CCZ14]. From the perspective of this note, the important point is that Gomory's fractional cut is the beginning of a cut-generating paradigm, not an isolated trick of the simplex tableau.

A second direction studies closures and their limits. The Chvátal closure is only one among several possible closure operators; split closures and mixed-integer closures can be stronger for mixed-integer sets, while still retaining a close connection to tableau reasoning [CKS90, MW01, ACL05]. These closures are central in modern branch-and-cut implementations, where solvers combine Gomory mixed-integer cuts, MIR cuts, cover cuts, clique cuts, and problem-specific separators [PR91, Ach09, ABCC06]. The practical lesson is that no single closure dominates all instances. Good solvers rely on families of cuts whose strengths are complementary.

A third direction asks how large an LP or SDP formulation must be if one is not allowed to generate cuts adaptively. Yannakakis' factorization theorem connected extension complexity with nonnegative rank [Yan91], and later work proved strong lower bounds for classical combinatorial polytopes, including the TSP polytope and the matching polytope [FMP+12, Rot14]. These results clarify why cutting planes remain useful even when finite descriptions exist. The finite description may be too large, and an adaptive cutting procedure may be the only reasonable way to expose the inequalities relevant to the current objective.

Finally, the hierarchy viewpoint connects cutting planes to proof complexity and approximation lower bounds. Sum-of-squares and semidefinite hierarchies can be seen as systematic ways to strengthen relaxations, and their lower bounds show that many natural relaxations require many rounds before they capture the relevant integer structure [Gri01, Lau03, Sch08, LRS15]. This literature should be read together with cutting-plane lower bounds [CKS90, Pud97], because both ask the same conceptual question in different proof languages. How many rounds of a restricted inference rule are needed before the discrete object is forced by the continuous relaxation?

8 Open Problems

The cutting plane method has been an active research area since Gomory's original papers, and several fundamental questions remain open. We discuss five problems spanning combinatorial rank bounds, proof complexity, and the computational efficiency of the method.

Open Problem 8.1 (Chvátal rank of the TSP polytope). The traveling salesman polytope TSP_n is the convex hull of characteristic vectors of Hamiltonian cycles in the complete graph K_n. Its natural LP relaxation adds the subtour elimination constraints, requiring that for every proper nonempty subset $S \subsetneq V$, the number of selected edges crossing the cut $(S, V \setminus S)$ is at least 2. This relaxation is one of the central test cases for comparing polyhedral strength with algorithmic usefulness. It is strong enough to support powerful approximation algorithms and separation routines, but it still admits fractional extreme points, and the precise number of CG rounds needed to eliminate them is poorly understood [DFJ54, HK70, Chr76, Ser78]. ⌟

The question asks for the Chvátal rank of TSP_n as a function of n. It is not asking whether the subtour relaxation is exact, because it is not; rather, it asks how many rounds of a very specific and canonical rounding procedure are needed before all subtour-fractional solutions disappear. General finite-rank theorems imply only that the process eventually terminates, and they do not distinguish between polynomial, quasipolynomial, or exponential behavior for this family.

A polynomial rank bound, say $r(\mathsf{TSP}_n) = n^{\mathcal{O}(1)}$, would show that a bounded number of closure rounds already captures the integer hull at the level of rank. This would not by itself give a polynomial-time algorithm, because each closure may contain exponentially many cuts and separation remains an additional issue. Conversely, an exponential lower bound would demonstrate that CG rank itself fails to explain the strength of the subtour relaxation in a small number of rounds.

Open Problem 8.2 (Exact rank of the matching polytope). The matching polytope MATCH_G of a graph G is the convex hull of characteristic vectors of matchings. Its LP relaxation, the fractional matching polytope, imposes only degree constraints $\sum_{e \ni v} x_e \leq 1$ and non-negativity. Edmonds' description of the integer matching polytope adds the odd-set inequalities, which require $\sum_{e \in E(S)} x_e \leq (|S| - 1)/2$ for every odd set $S \subseteq V$ [Edm65b, Edm65a]. ⌟

For bipartite graphs, the degree relaxation is already integral, which is the polyhedral shadow of Kőnig's theorem [Kőn16]. For non-bipartite graphs, the odd-set inequalities are necessary, and the rank question asks how quickly CG derivations recover them from the fractional relaxation. In the complete graph, does one round already derive all odd-set inequalities from the degree constraints and non-negativity, or are further closure rounds genuinely needed?

An answer would clarify the relation between separation and derivation. Edmonds' algorithm can separate over odd sets efficiently, but efficient separation does not automatically imply low CG rank from a fixed starting relaxation. The problem therefore asks whether the combinatorial simplicity of odd-set separation has a parallel proof-theoretic explanation.

Open Problem 8.3 (Proof complexity of cutting plane refutations). The cutting-plane proof system [CKS90] is a propositional refutation system where axioms are 0-1 linear inequalities encoding a CNF formula F and the derivation rule is exactly the CG step, deriving $\mathbf{a}^\mathsf{T}\mathbf{x} \leq \lfloor b \rfloor$ from $\mathbf{a}^\mathsf{T}\mathbf{x} \leq b$. A CP refutation of the unsatisfiability of F is a derivation of $0 \geq 1$. CP polynomially simulates Resolution, so it is at least as strong [CKS90, Hak85]. ⌟

Pudlák [Pud97] proved lower bounds for cutting-plane proofs using methods that connect linear inequalities with monotone computation. The general lesson is that CP is much stronger than Resolution, because it can manipulate large integer coefficients, but this strength also makes lower bounds harder. Width-based arguments that work for Resolution do not transfer directly, since a single CP line may encode a global counting statement [Hak85, CS88, BSW01].

The concrete question is whether one can prove superpolynomial CP lower bounds for the unsatisfiability of a random 3-CNF formula near the satisfiability threshold $m/n \approx 4.27$. For Resolution, the size-width method of Ben-Sasson and Wigderson [BSW01] gives a robust route to exponential lower bounds. For CP, the long integer coefficients that cuts may use defeat the width argument. Establishing such lower bounds would give a much sharper picture of what linear-inequality reasoning can certify about random constraint systems, and it would separate CP from several weaker proof systems on a natural distributional family. It would not, by itself, separate CP from Extended Frege, because Extended Frege may have short proofs even when CP does not.

Open Problem 8.4 (Gomory cuts versus lift-and-project). Lift-and-project cuts [Bal71, NW88, BCC93] are generated by a different mechanism. One multiplies the LP constraints by a single variable x_j (or $1 - x_j$ for 0-1 variables) to lift the problem to a higher-dimensional space, then projects back to the original space. For 0-1 programs, one round of lift-and-project recovers all valid inequalities derivable from a single variable. Several related hierarchies make the same basic tradeoff between higher-dimensional relaxations and stronger projected inequalities [SA90, LS91, Las01]. ⌟

The separation question asks whether there are families of polytopes where Gomory's fractional cuts require polynomially or exponentially more rounds than lift-and-project to reach P_{I}, or vice versa. The two methods are incomparable on specific families. Lift-and-project handles bilinear terms naturally through the disjunction $x_j \in \{0, 1\}$, while Gomory cuts may be stronger for problems without binary structure. A clean theoretical separation, meaning a family of polytopes where the Gomory rank is $\omega(1)$ but the lift-and-project rank is $\mathcal{O}(1)$ or vice versa, would clarify the relative power of the two approaches and guide solver designers.

Open Problem 8.5 (Optimal cut selection). At each iteration, the Gomory algorithm may choose any fractional row i to generate a cut. Different choices lead to different sequences of LP relaxations and potentially very different total cut counts. In practice, commercial solvers implement heuristic selection rules (deepest cut, most violated, highest violation density), but no selection strategy is known to guarantee polynomial convergence. ⌟

The algorithmic question asks whether there is a polynomial-time cut selection strategy such that the Gomory cutting plane algorithm terminates in polynomially many iterations for any fixed-dimension IP. In fixed dimension n, Lenstra's algorithm [Len83] already solves IP in polynomial time, but by a fundamentally different method, namely lattice basis reduction rather than cutting planes. A polynomial cut-selection scheme would unify the two approaches and might yield a practically competitive algorithm. Even understanding the complexity of the optimal sequence length, that is, the minimum number of Gomory cuts needed to reach P_{I} from any initial LP basis, would be a significant advance.

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